Murphy Bookshelf Door

Build design, buying list and responsibility for the measurements

Design: a bookshelf that opens outward into the hallway. Standing in the hall facing the closet, the pivot is on the left, toward the bathroom. Press the free edge to release the latch; the door pops outward, then pull it open.

1. What is settled, and what is a build proposal

Item	Basis / status
Location and swing	User-marked closet on the right below the bathroom; original design log records left pivot / hall outswing. Full-depth swing clearance still needs a site mockup.
Size	Recorded nominal door: 36 in wide x 80 in high. Recorded closet: 36 x 83 in; this is not a verified finished opening.
Construction selected here	Paint-grade 3/4 in plywood carcass, 4 fixed shelves, applied 1/4 in plywood back, 3/4 in face frame and shelf-front lips. Shelf count and joinery are this plan's proposal.
Depth selected here	8-1/4 in total, including 1/4 in back and 3/4 in face frame. Shelf panels are about 7-3/16 in deep. For 8 in usable shelves, use a 9 in total case and a new pivot mockup.

2. Hardware to buy

Qty	Item	Budget
1	Murphy Door Single Hidden Door Hardware Kit: top/bottom pivots, plates and supplied fasteners. Follow left-hand outswing layout.	$150
1	Sugatsune ML-ZN80-BL full-size-door push-to-open latch, black; strike and fasteners supplied. Replaces the friction catch for this design.	$53
1	Padded door stop, location selected after the sweep mockup.	Price open
	Known hardware subtotal; tax, shipping and other materials additional.	$203

Order links: Murphy hinge kit | Sugatsune ML-ZN80-BL latch
Listed prices checked October 10, 2026. No purchase has been made.

Field Measurement Sheet

Crew completes these values; Tisa does not need to guess them

Before measuring: identify the finished floor elevation and final jamb face. Label each photo with the measurement and pivot side. All dimensions are inches unless noted.

Measure / check	Record actual values
Finished clear opening width	Top W1 _____ Middle W2 _____ Bottom W3 _____ Smallest W = _____
Finished clear opening height (from final floor)	Left H1 _____ Middle H2 _____ Right H3 _____ Smallest H = _____
Opening square / plumb / level	Diagonal A _____ Diagonal B _____ Jamb plumb deviation _____ Floor level deviation _____
Jamb and wall	Jamb depth _____ Wall thickness _____ Jamb material _____ Structural blocking location _____
Pivot-side obstruction	Bathroom-side wall / casing projection _____ Adjacent bathroom door sweep or other interference _____
Outward sweep / hall passage	Hall width _____ Case front to opposite obstruction _____ Full-depth cardboard sweep passes: YES / NO Clear passage with bookshelf fully open _____
Appliance behind door	Exact make / model _____ Back of closed case to appliance _____ Service / removal route with case open _____
Air openings and utilities	Manual-required free air area / opening locations _____ Duct, gas, electrical and shutoff access retained: YES / NO
Final material thickness	Plywood t _____ Back b _____ Face-frame f _____ Actual sheet size _____ Saw kerf _____
Verified hardware geometry	Pivot offsets copied from correct kit template and mockup: _____ Width reduction Aw _____ Height reduction Ah _____

Release before final cuts: measurer _____ installer _____ date _____
Attach photos of W1/W2/W3, H1/H2/H3, diagonal checks, full hallway sweep and the appliance label. The appliance record conflicts between LG and Samsung; use the actual delivered model.

Cut List and Formulas

Worked example is for estimating; final cuts come from the site measurements

NOT RELEASED FOR CUTTING: the example uses a 36 x 80-7/8 in finished jamb, with illustrative reductions of 1-1/4 in width and 2-3/8 in height. These reproduce a vendor example; they are not universal allowances and are not the site measurement.

Set: Wc = smallest finished width - Aw; Hc = smallest finished height - Ah.
D = overall case depth; t = carcass thickness; b = applied back thickness; f = face-frame thickness. Panel depth P = D - b - f. Shelf depth S = P - 1/16 in rear relief.

Qty	Part / material	Final formula	Example inches
2	Sides / 3/4 in plywood	Hc x P	78-1/2 x 7-1/4
2	Top and bottom / 3/4 in plywood	(Wc - 2t) x P	33-1/4 x 7-1/4
4	Fixed shelves / 3/4 in plywood	(Wc - 2t) x S	33-1/4 x 7-3/16
1	Applied back / 1/4 in plywood	Wc x Hc	34-3/4 x 78-1/2
2	Face-frame stiles / 3/4 in stock	Hc x 1-1/2	78-1/2 x 1-1/2
2	Face-frame rails / 3/4 in stock	(Wc - 3) x 1-1/2	31-3/4 x 1-1/2
4	Shelf-front lips / 3/4 in stock	(Wc - 3) x 1	31-3/4 x 1
1+1	Top and bottom pivot blocking	Size to approved hardware / mockup	Field sized
Field	Jamb, casing, fixed latch block	Measure on site; preserve push travel	Field sized

Example assembled case: Wc = 34-3/4; Hc = 78-1/2; D = 8-1/4; t = 3/4; b = 1/4; f = 3/4. Side panels span from the rear of the face frame to the front of the applied back. Shelves leave 1/16 in at the rear; front lips sit in the face-frame thickness zone.

Shelf spacing: interior height I = Hc - 2t. With 4 evenly spaced shelves, each clear carcass bay B = (I - 4t) / 5. Example: I = 77; B = 14.8 in. The 1 in lips project 1/4 in above each shelf, reducing the front access opening. Mark shelf positions from the bottom panel's top, not from the floor.

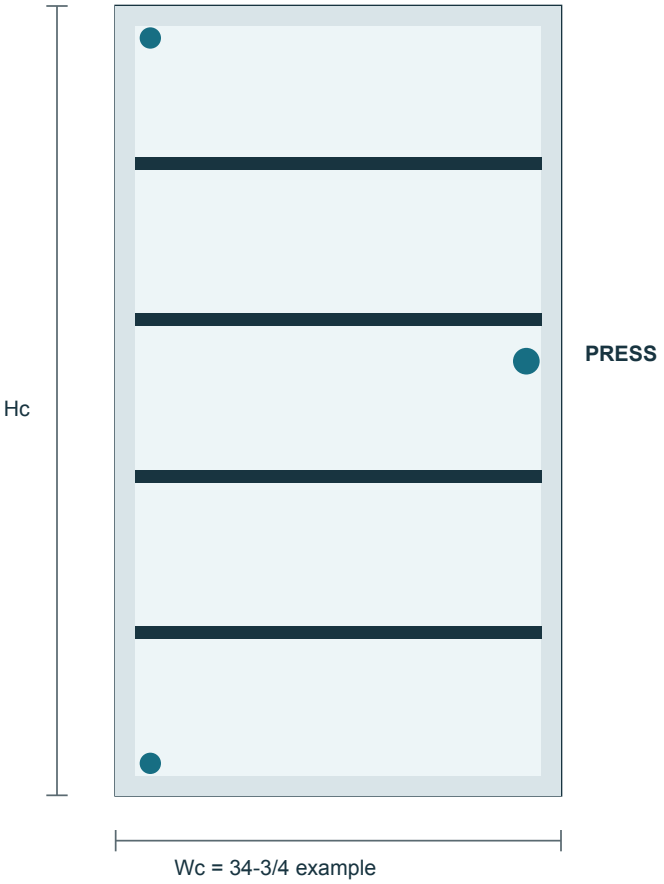
Shelf bottom elevations above bottom panel top: B; 2B + t; 3B + 2t; 4B + 3t. Example: 14.8, 30.35, 45.90, 61.45 in. Transfer these from one story stick; do not accumulate rounding.

Design Drawings

Own schematic views; not a drilling template or field-approved shop drawing

A. FRONT - hallway view

Bathroom side / pivot LEFT



4 fixed shelves; lips shown dark

Two pivot marks indicate side only.

Actual positions follow kit / mockup.

B. PLAN - top view

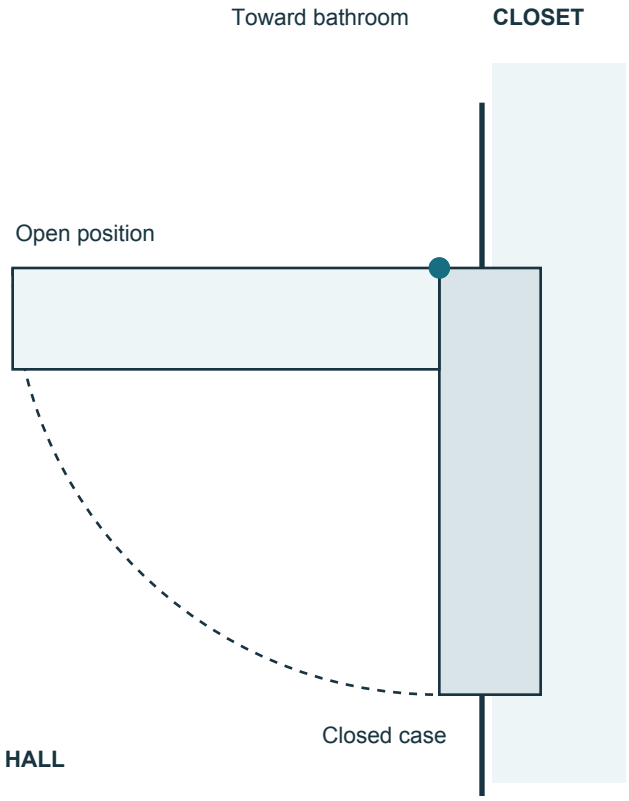
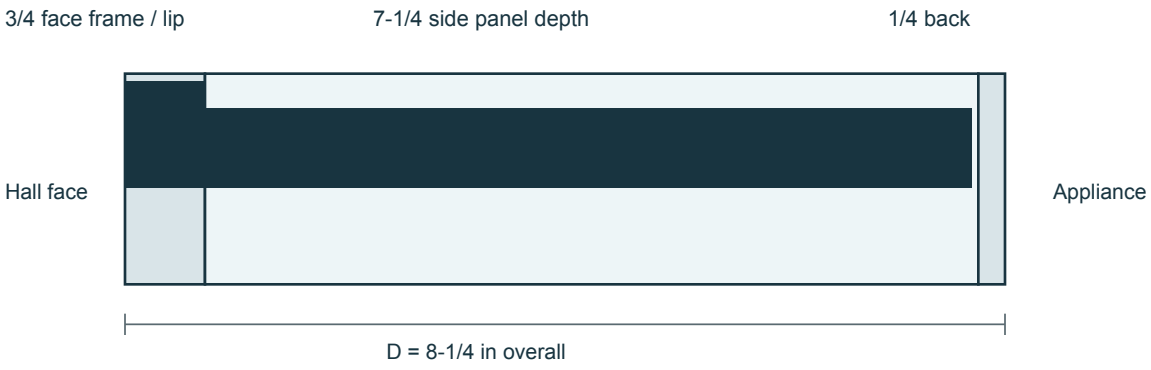


Diagram illustrates outward swing only. Use the full case depth and actual pivot offsets for the cardboard sweep. Check nearby bathroom / bedroom doors and hall passage.

C. SECTION - depth stack



8 in usable shelf depth requires a deeper case (approximately 9 in overall with this stack), different swing verification and adjusted pivot layout.

Build and Install Sequence

Numbered tasks - check the stock and measure before buying duplicate materials

1. Measure and check the stock (1-2 person-hours). Complete page 2, identify final floor height, inspect existing plywood / trim / fasteners, and read the actual appliance installation manual. Identify required closet air openings and service clearances.
2. Make a full-depth mockup (1 person-hour). Use cardboard or scrap to represent the whole case, not a thin door. Test the bathroom-side pivot, hall sweep and removal path. Apply the actual left-outswing kit geometry; record each pivot offset. Both hinge axes must line up vertically.
3. Release the cut sizes and cut the parts. Enter actual values in the linked calculator. Verify material thickness and saw kerf. Mark the two sides together with a story stick; retain matching parts. Keep pivot and latch blocking on the site-specific cut list.
4. Dry-fit, then assemble square. Top and bottom go between the full-height sides. Use cabinet joinery and predrilled assembly fasteners appropriate to the actual stock. Fix all 4 shelves. Check diagonals, then glue and screw the applied back to restrain racking. Add the face frame and lips; ease exposed edges.
5. Reinforce and fit the hardware. Provide solid fastening at pivot and latch zones without blocking the kit's pin access or sweep. Use the supplied pivot screws and the correct left-outswing layout. Do not substitute guessed drill positions from this schematic.
6. Prepare the jamb; hang with two people. Anchor into sound structural framing with site-selected anchors. Set the jamb square and plumb at finished floor elevation. Dry-fit the bookcase; verify full pin engagement and upper/lower alignment. Check swing before finishing the trim.
7. Fit the touch latch and stop. Mount the latch on fixed blocking opposite the pivot and its strike on the case rear. Leave the specified 3 mm inward press travel; the spring then extends about 25 mm. Trim and stops must permit this motion. Do not fit the resisting friction catch in parallel. Add a padded stop after checking the complete sweep.
8. Finish, ventilate and test. Fill/sand, prime and finish all edges. Let coatings cure before cycling the latch. Fit any manual-required air openings while preserving case stiffness. Test release and swing with the intended contents, then deliver photos and the signed checks on page 7.

Planning labor: measure + mockup 2-3 person-hours; fabrication 6-10; hanging / adjustment 4-6; finish 3-5. Total: 15-24 person-hours, with glue/paint curing separate. One working day is a target for a prepared assembly's installation, not a promise to measure, fabricate, cure and install from scratch.

Materials and Hardware Details

Quantities for the selected design; check existing materials first

Qty / allowance	Material	Status / selection
1 sheet	3/4 in paint-grade cabinet plywood, 4 x 8 ft	Worked example carcass / shelves fit 5 strips, each 7-1/4 in wide. New field dimensions may increase quantity.
1 sheet	1/4 in plywood, 4 x 8 ft	Applied back; cut as one piece where practical.
4 boards	1 x 2 x 8 ft paint-grade poplar (actual 3/4 x 1-1/2 in)	Two 78-1/2 in stiles from two boards; two more boards yield two rails + four lips, each 31-3/4 in long. Rip lips to 1 in height.
1 bottle	Wood glue, approximately 16 oz	Check stock; follow its cure time.
1 box	Cabinet assembly screws, approximately 100	1-1/4 in planning size for appropriate assembly locations; confirm joinery, pilot holes and penetration.
Field sized	Structural jamb anchors / reinforcing stock	Type, length and count depend on framing, substrate and kit geometry; installer records them before release.
1 + 1 + 1	Pivot kit + ML-ZN80-BL latch + padded stop	One of each. Kit includes pivot fasteners; latch includes strike / fasteners.
Field established	Primer, paint, filler, edge treatment	Check available stock and coverage. Finish color remains a Tisa decision.
Manual dependent	Ventilation grilles / free-air openings	Actual gas dryer model and manual decide size / location. A solid unvented back is not released for use until this is resolved.

Example plywood nesting: rip 5 strips at 7-1/4 x 96 in. Two yield the sides (78-1/2 long); three yield two boards each (33-1/4 long), for top, bottom and 4 shelves. Trim the shelf depths to 7-3/16 in. Allow the actual kerf; 5 strips with 1/8 in rip kerf use under 37 in of sheet width. Keep offcuts for mockups / blocking where suitable.

Load and latch checks

Design for the lower applicable component reference: assembled case plus contents no more than 275 lb with the selected touch latch. The Murphy kit manual states 300 lb while its product page states 400 lb; do not use the higher claim for this plan. The latch reference is not a certified load rating for the combined DIY system.

The pivots support the case; the touch latch retains and starts the opening motion. Its 6.6 lb magnetic hold is not a weight-support rating. Test the chosen location, blocking, trim gap and contents with the real case. Provide an accessible way to open the door from inside; do not add a keyed lock to this laundry design.

Hardware envelope and adjustment

Touch latch: envelope approximately 2.44 in closed / 3.43 in extended, 1.54 in wide, 0.87 in high. Mount on solid blocking; establish strike spacing from its drawing. Leave 3 mm press travel, separate from the casing reveal.

Pivot kit: use its left-outswing drawing at verified print scale. Custom depth, width, jamb and trim alter offsets: validate with the full-depth mockup. Drilling centers are not released until these are set.

Completion and Crew Handoff

Work is complete only when the door operates and the laundry remains accessible

Completion checklist

Check	Record / evidence
Measurements / final cuts	Page 2 complete; calculator and final cut sizes saved. Installer _____ date _____
Case and mount	Case square; back fixed; reinforced anchors; pins fully engaged. Photo _____
Press and swing	Releases after the 3 mm press; opens without binding or collision. Empty test _____ loaded test _____
Contents and passage	Case + contents weighed / estimated conservatively _____ lb. Retention lips fitted; hall route and appliance removal checked.
Laundry ventilation / service	Actual model _____. Required air area _____. Grilles / service / gas, duct and electrical access documented _____
Finish / stop / interior opening	No rubbing after cure; padded stop fitted; inside opening works. Photos _____

Resumen de trabajo para el equipo

Puerta librero del clóset de lavandería: abre hacia afuera, al pasillo. Viéndola desde el pasillo, el pivote va a la izquierda, del lado del baño. Presionar el borde libre libera el cierre; luego se jala la puerta.

1. Medir el ancho en tres puntos y la altura en tres puntos desde el piso terminado; tomar las dos diagonales.
2. Hacer una plantilla del librero completo, incluyendo su profundidad; revisar que no choque con otras puertas.
3. Confirmar el modelo real de la secadora y sus requisitos de ventilación antes de cerrar el clóset con un panel sólido.
4. Usar las medidas reales para el corte. 36 x 80 pulgadas es nominal; no es la medida de corte.
5. Instalar pivotes, cierre y tope. Dejar 3 mm de movimiento hacia adentro para liberar el cierre.
6. Entregar fotos y probar la puerta con su carga prevista. No comprar material duplicado sin revisar lo disponible.

Live working records and manufacturer references

Measurements / cut calculator

Master scope - Murphy Door Shelf tab

Murphy kit installation manual

Sugatsune ML-ZN80 specification / 3 mm release stroke

Source basis: October 7 design log; Tisa's attached floor plan and swing direction; official manufacturer product pages and manuals. Site measurements are pending. Combined Murphy pivot / Sugatsune latch installation is a proposed custom integration, not manufacturer-certified. Hardware prices exclude tax and shipping.

Code and access requirements

Governing addendum - October 12, 2026 | Original seven pages retained

Read these two pages before ordering grilles, cutting the case or closing the laundry doorway. They govern earlier generic model, air-opening and passage notes, and the original pricing/order-status snapshot. They do not release the example cuts.

1. Codex owns the verification

Codex checks the applicable permit/code edition, the exact appliance manual, the air-path design, the hardware geometry and the actual installation evidence. The crew supplies labeled measurements, appliance-label photos and installation photos. Tisa does not need to research the code herself. Current installation compliance is **not yet verified**.

The 2025 California Mechanical Code is the current state baseline, effective January 1, 2026, with San Francisco amendments. SFBC 101A.4.1 ties the governing edition to the permit accepted for review. Public records found a Unit 2 permit, not an unambiguous current Unit 4 permit; Codex still owns that code-basis check.

2. The laundry closet must have makeup air

CMC 504.4.1 requires a dryer-closet air opening of at least **100 square inches** in the door, or another approved means, plus the exact dryer manufacturer requirements. Do not seal the bookshelf and assume the trim gaps supply enough air.

- Codex specifies the air path and any high/low opening locations after confirming the actual appliance. John measures; the crew installs the specified openings and photographs them.
- **Use net free area, not the grille outside dimensions.** If an opening needs 100 sq in free area and its grille is 60% free, its gross opening must be at least 100 / 0.60 = 166.7 sq in. Manufacturer requirements can call for more or differently located openings.
- Do not choose or cut grille holes until the verified free-area data and appliance manual are recorded. Keep air paths free of books, the case back and stored items.

3. Prove that the appliance can pass through

Murphy Door states that a standard **8.25 in deep outswing** reduces the opening by **11.25 in**. A 36 in jamb could therefore leave only **24.75 in** in that standard configuration. This is a manufacturer example, not the measured custom-door passage.

- Measure the clear passage with the full-depth case open, including the pivot, trim and stops. Compare it with the exact appliance removal envelope and required access.
- If the appliance cannot pass, design and document a removable-bookcase procedure or adjust the case geometry before cutting. Room to turn inside does not solve a narrow doorway.

If the ordered LG WKG101HVA remains the selected machine: its 27 in body is wider than the 24.75 in standard-door example. A verified custom passage or removable-bookcase procedure is required; do not treat the original example as an appliance-access approval.

Code: 2025 CMC, section 504.4.1 (viewer page 147) | Murphy Door FAQ: inswing vs outswing | SFBC 101A.4.1

Hardware ordered; latch to buy

Governing addendum - October 12, 2026 | Door remains a custom integration

4. Actual order and remaining purchase

Qty	Exact item / purchase link	Price
1	Murphy Door Single Hidden Door Hardware Kit (FMHD) **ORDERED - Order 100070, Oct. 12, 2026, 9:07 p.m. PDT.** Item $150 + shipping $20 + tax $12.94. No arrival date confirmed. Use left-pivot / outswing template.	**$182.94** Actual paid
1	Buy Sugatsune ML-ZN80-BL latch at PlumbTile **NOT ORDERED.** Black ML-ZN80-BL selected; quantity 1. Retailer has an Add to Cart checkout. Use its strike and supplied fasteners.	**$45.05** Before its tax/shipping
	Actual kit spend + planned latch price. Only $182.94 has been paid. The latch purchase remains pending.	**$227.99** Plus latch tax/shipping

Latch retailer/price rechecked October 12, 2026. Sugatsune's own page provides specifications and a Where to Buy route, not direct checkout. This addendum supersedes the original $203 estimate and no-purchase statement. Check existing stock before buying build materials or a stop; grille and appliance-vent selections remain unreleased.

5. How this hardware will operate

- **Hallway view:** pivot on the left, toward the bathroom; case opens outward into the hall. Press the free edge inward to release, then pull outward.

- Sugatsune specifies **3 mm inward release travel**. Trim, stops and blocking must permit it. Its spring projects about 25 mm after release. Do not install the resisting friction catch at the same time.

- Use the Murphy kit manual's conservative **300 lb** reference, rather than its higher product-page claim. Sugatsune's **275 lb door reference** is not a support rating or certification of this mixed-brand bookshelf assembly. Verify case-plus-contents load, anchors, release force and loaded cycling; the lower reference governs planning.

- Confirm full-depth swing, vertically aligned pivots, exact template offsets and inside opening before final cuts. A change to 8 in usable shelf depth requires **9.0625 in overall depth** under this plan's back/frame/rear-relief stack; rerun the mockup and cuts.

6. Confirm which laundry appliance is actually coming

Records show an **LG WKG101HVA** September 4 receipt, September 27 reschedule for October 31 delivery and October 8 payment failure. Tisa later identified **Samsung**. Delivery/installation remains unverified. Codex matches the actual appliance label/current order to its manual before releasing grilles, clearances or passage.

Conditional LG reference only: manual MFL71996109 Rev06_092926 lists a 27 in wide x 74-3/8 in high x 30-1/4 in deep body; minimum clearances are 1 in at the sides/top and 4 in at rear/closet front. Its closet instructions require an upper opening of at least 48 sq in, at least 6 ft above the floor, and a lower opening of at least 24 sq in, no more than 1 ft above the floor; the diagram shows 3 in from door top/bottom edges. Apply the CMC 100 sq in total floor as well as these model-specific locations. Final grilles remain unreleased until Codex confirms the current appliance. See manual pages 13-14.

LG WKG101HVA official support | LG installation manual

Murphy kit installation manual | Sugatsune manufacturer specs | Latch dimensions / release stroke | Live measurement form